

Health App Fitness + Nutrition v0.2

YouTube Embeds, AI Import, Calendar Activation and History Implementation Plan

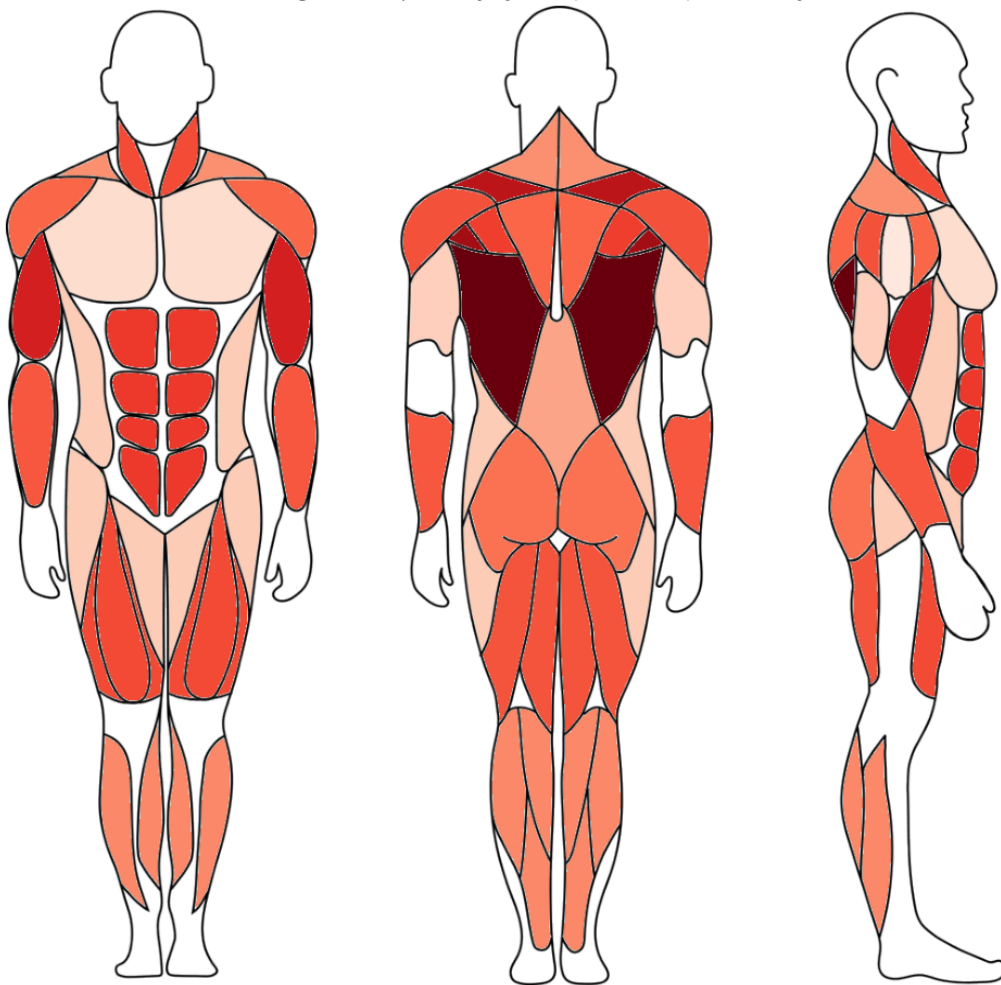
Generated for Keeno Chetty - 2026-06-11

Purpose

This document explains how to extend the fitness content seed pack so the app can load exercises, nutrition templates, diet styles, workout plans, YouTube demo embeds, AI-imported plans, calendar events and user history from the database.

The safest design is: app-owned exercise instructions and muscle maps as the core content, with YouTube as an optional embedded demo layer. AI imports should become editable drafts first, not instant medical or training advice.

Muscle Usage Heat Map - Decay by Time (Factor: 0.25) - Last 7 Days



Reference visual direction: dynamic muscle-map heat layers. Production should use reusable body-map layers rather than thousands of static images.

1. What was added to the ZIP

Area	File(s)	Use in app
YouTube video embeds	data/exercise_video_sources_seed.csv/json	Candidate metadata table for official YouTube embeds.
Exercise categories	data/exercise_type_taxonomy.csv	Filters for strength, yoga, cardio, running, pregnancy, child, teen and advanced training.
Diet categories	data/diet_type_taxonomy.csv	Diet support for keto, vegan, vegetarian, carnivore, Mediterranean, medical-gated styles and more.
AI import flow	data/ai_import_workflow_seed.json	Defines search, normalize, review, preview, import, activate and history steps.
Calendar activation	data/calendar_activation_rules.csv	Rules for scheduling workouts, meal reminders and recovery blocks.
History	data/history_event_taxonomy.csv	Events to store imports, activations, completed sessions, edits and warnings.
Database migration	sql/002_youtube_ai_import_schema.sql	Supabase tables and columns for the new functionality.
Codex prompt	docs/CODEX_PROMPT_fitness_youtube_ai_import.md	Implementation prompt for your health app codebase.

2. Database architecture

The app should load baseline content from global tables and user-customized content from user tables. This lets you keep one official exercise library while each user can import, edit, share, pause or delete their own plans.

Layer	Tables	Purpose
Global library	fitness_exercises, fitness_workout_programs, fitness_nutrition_templates	Approved seed exercises, plans and nutrition templates.
Media layer	fitness_exercise_video_sources, content_licenses	YouTube metadata, source links, attribution and review status.
Taxonomy	fitness_exercise_type_taxonomy, fitness_diet_type_taxonomy	Filtering, personalization and safety gates.
User imports	user_imported_plans, user_imported_plan_days	AI-imported or user-created plans stored per user/family member.
Calendar	user_plan_calendar_events	Scheduled workout, recovery, nutrition and health reminders.
History	user_fitness_history, ai_plan_search_logs	Audit trail of what the user searched, imported, changed and completed.

3. YouTube integration

Use YouTube as an external embedded demo provider. Store only metadata such as video ID, title, channel, thumbnail URL, embeddable flag, license type and review status. Do not download or package video files.

Recommended search filters for candidate discovery: type=video, videoEmbeddable=true, videoSyndicated=true, safeSearch=strict, and optionally videoLicense=creativecommons. Every result should still go to manual review before public display, especially for pregnancy, children and advanced training.

UI placement: exercise detail page shows app-owned steps, muscle map, warnings and then an optional "Watch demo" YouTube embed. If the video is removed or blocked, the exercise must still work without it.

4. AI search and import flow

Step	Action	Database result
1	User searches for a goal, diet or workout not in the library.	ai_plan_search_logs
2	Backend searches approved web sources and returns source URLs.	sources_found json
3	AI normalizes the result into your schema using app-owned wording.	draft user_imported_plans
4	Safety/license gate checks audience, pregnancy, child, medical, restrictive diet and advanced flags.	license_status, safety_flags
5	User sees preview, edits days/times/exercises/meals and confirms.	preview_ready/user_edited
6	User clicks Import and Activate.	user_imported_plan_days + calendar events
7	App tracks completions, swaps, missed sessions and edits.	user_fitness_history

5. Safety gates

Audience/plan	Gate
Pregnancy	Trimester-aware warnings, no auto-extreme plans, clinician clearance language, stop signs.
Postpartum	Return-to-exercise clearance, pelvic floor/core symptoms, gentle progressions.
Baby/child	Pediatric review, parent supervision, no adult dieting or intense plans.
Teen	Parent/coach guidance, food-first nutrition, supplement warnings.
Keto/carnivore/low FODMAP/diabetes/DASH	Medical/dietitian review flags where relevant.
Extreme/insane training	Advanced-only gate, recovery spacing, RPE tracking and explicit warnings.

6. App screens to build

Screen	Main components
Fitness Home	Today workout, active plan, recovery score, muscle heat map, quick log.
Exercise Library	Filters by type, goal, equipment, audience, level, muscle, safety and video availability.
Exercise Detail	Instructions, sets/reps, progressions, regressions, muscle map, warnings, YouTube embed.
Plan Library	28-day plans, goal progressions, pregnancy, child, teen, nutrition and recovery templates.
AI Plan Search	Search box, filters, source list, safety/license result and import preview.
Import Preview	Calendar preview, days, sessions, meals, reminders, edit/swap controls and warnings.
Active Plan	Progress, next session, pause, edit, share, duplicate, archive/delete.
History	Completed sessions, skipped sessions, edits, imports, nutrition logs, warnings and AI searches.

7. Implementation phases

Phase	Build
1	Run schema and import seed/taxonomy data.
2	Build read-only exercise library and nutrition template browser.
3	Add YouTube candidate search for admins and video review approval.
4	Add exercise detail page with optional official YouTube embed.
5	Build user-created/imported plan tables and plan editor.
6	Build calendar activation: start date, preferred days, time slots and reminders.
7	Build history logging and progress views.
8	Add AI web search import with source/license/safety preview.
9	Add sharing, duplicate, pause, archive and delete controls.

References and policy notes

- YouTube Data API search.list supports videoEmbeddable, videoSyndicated and videoLicense filters for video search candidates.
- YouTube IFrame Player API is the correct route for embedded playback and player controls.
- YouTube API policies should be followed: do not download, store or make YouTube audiovisual content available offline without permission.
- Use professional review before publishing pregnancy, baby, child, teen, medical diet, supplement or extreme training content.